

Firebase Integration — Lava

This document describes how Firebase is integrated into Lava (Android client + lava-api-go service) and how to operate the distribute / stats workflows.

Sensitive values are NEVER documented here. All real tokens, project IDs, app IDs, and tester emails live in `.env` (gitignored). The placeholders in `.env.example` are illustrative only.

Services enabled

Service	Where	Purpose
Crashlytics	Android client	Fatal + non-fatal exception reporting; custom keys for <code>build_type</code> , <code>version_name</code> , <code>version_code</code> , <code>application_id</code>
Analytics	Android client	User-visible flow events: login, search, browse, view-topic, download-torrent, provider-selected, endpoint-discovered
Performance Monitoring	Android client	Auto-instrumented HTTP traces; custom traces wired via <code>Trace.startTrace()</code>
App Distribution	Android (debug + release APKs)	Tester invitations driven by <code>.env</code> roster

Setup (one-time)

1. **Authenticate Firebase CLI:** `firebase login:ci` — copy the printed token into `.env` as `LAVA_FIREBASE_TOKEN`.
2. **Create a Firebase project + apps** (already done for Lava; project ID = `lava-vasic-digital`). For a fresh fork:

```

firebase projects:create your-project-id --display-name "Your
App"
firebase apps:create ANDROID --project your-project-id --
package-name your.package.name
firebase apps:create WEB --project your-project-id "Your Web
App"

```

Copy the resulting app IDs into .env under
LAVA_FIREBASE_ANDROID_APP_ID and LAVA_FIREBASE_API_GO_APP_ID.

3. Refresh local config files:

```
./scripts/firebase-setup.sh
```

This re-fetches app/google-services.json (Android) and lava-api-go/firebase-web-config.json (Web). Both are gitignored.

4. Add tester emails to .env: LAVA_FIREBASE_TESTERS_OWNER, LAVA_FIREBASE_TESTERS_DEVELOPER, LAVA_FIREBASE_TESTERS_TESTER. The list is loaded fresh each time firebase-distribute.sh runs.

Android client wiring

- **Plugins** applied via buildSrc/.../
AndroidApplicationConventionPlugin.kt:
 - com.google.gms.google-services
 - com.google.firebase.crashlytics
- **Dependencies** in gradle/libs.versions.toml:
 - firebase-bom (33.15.0) — version-aligns the SDKs below
 - firebase-analytics
 - firebase-crashlytics
 - firebase-perf
- **Bootstrap** in app/src/main/kotlin/digital/vasic/lava/client/
LavaApplication.kt:
 - FirebaseApp.initializeApp(this) — must be first to use the SDKs
 - Crashlytics: collection enabled in release builds only; custom keys (build_type, version_name, version_code, application_id) attached
 - Analytics: collection enabled in release builds only; user properties (build_type, app_version) set
 - Performance: collection enabled in release builds only
- **Tracker abstraction** at core/common/src/main/kotlin/lava/common/analytics/AnalyticsTracker.kt — Hilt-injectable interface with canonical event names. Features depend on this interface; the Firebase implementation lives in :app (digital.vasic.lava.client.firebase.FirebaseAnalyticsTracker) and is bound via FirebaseModule.kt. This keeps the Firebase coupling in :app per the Decoupled Reusable Architecture rule — features remain reusable.

Distribution workflow

The canonical operator distribute flow (replaces the old releases/-only delivery):

```
# Rebuild + distribute everything
./scripts/distribute.sh

# Variants
./scripts/distribute.sh --no-rebuild      # use existing
                        artifacts
./scripts/distribute.sh --android-only    # rebuild +
                        distribute Android only
./scripts/distribute.sh --release-only    # only the release-
                        signed APK
./scripts/distribute.sh --debug-only      # only the debug APK
```

Internally:

1. `./build_and_release.sh` rebuilds all artifacts into releases/
<version>/.
2. `./scripts/firebase-distribute.sh` uploads the Android APKs to
Firebase App Distribution and notifies testers from `.env`.
3. `./scripts/firebase-stats.sh` prints the dashboard URLs the
operator uses to inspect crashes / events / performance.

Anti-bluff posture: every script uses `set -euo pipefail`. There is no `|| echo WARN` swallowing of failures. A failed APK upload propagates as a non-zero exit, breaking the distribute chain.

Stats / observability

Live dashboards (the URLs are project-public; the data is access-controlled):

- Crashlytics: https://console.firebase.google.com/project/<PROJECT_ID>/crashlytics
- Analytics: https://console.firebase.google.com/project/<PROJECT_ID>/analytics
- Performance: https://console.firebase.google.com/project/<PROJECT_ID>/performance
- App Distribution: https://console.firebase.google.com/project/<PROJECT_ID>/appdistribution

Quick CLI: `./scripts/firebase-stats.sh (--days N window, --json machine-readable)`.

For deeper queries, the `lava-api-go` service will expose `/admin/firebase-stats` (via the Firebase Admin SDK) — see `lava-api-go/internal/firebase/`.

Sensitive files (NEVER commit)

.gitignore lists:

- .env, .env.*, .env.local
- app/google-services.json, **/google-services.json
- lava-api-go/firebase-web-config.json
- lava-api-go/firebase-admin-key.json, firebase-admin-key.json, **/firebase-admin-*.json
- firebase-debug.log, **/firebase-debug.log
- .firebase/
- firebase-distribute-*.log

Pre-push hook + scripts/check-constitution.sh reject pushes that introduce these files (constitutional 6.H clause 5).

Rotation

If LAVA_FIREBASE_TOKEN is suspected leaked:

1. Revoke at <https://myaccount.google.com/permissions> (search "Firebase CLI").
2. firebase login:ci to mint a new one.
3. Replace in .env. Do not commit.

If firebase-admin-key.json is suspected leaked: revoke the service account in the Firebase Console → Project Settings → Service Accounts and create a new one. Replace in lava-api-go/firebase-admin-key.json. Do not commit.

Testing posture (anti-bluff)

Per constitutional clauses §6.J + §6.L (Anti-Bluff Functional Reality Mandate, invoked TEN TIMES on this project), Firebase tests MUST verify real reporting, not mocked calls:

- app/src/androidTest/.../FirebaseIntegrationTest.kt — installs the debug APK, triggers a known non-fatal, asserts the Crashlytics SDK accepted the report (collection enabled, no error from recordException).
- app/src/test/.../FirebaseAnalyticsTrackerTest.kt — verifies AnalyticsTracker calls Firebase SDK with the canonical event names + parameters.
- Falsifiability rehearsal recorded in commit body per §6.N: deliberately break event() (e.g., return without calling analytics.logEvent) → assert the test fails.

A Challenge Test (C9 onward) for end-to-end Firebase visibility — an operator-attested flow where a deliberate non-fatal in the debug build appears in the Crashlytics dashboard within 60 seconds — is the load-bearing acceptance gate.